

PAPER 1: MEDICINE THROUGH TIME WITH THE WESTERN FRONT

WESTERN FRONT

Assessment Question: How has medicine changed from c1250 to the modern day?

Name: _____

Class: _____

Progress & Assessment Tracker	Effort	Level	Teacher Comments	
L1: KT5.1: What was the historical and medical context of the Western Front?				
L2: KT5.2: How did the trench environment create new medical challenges?				
L3: KT5.3: What new illnesses and wounds did soldiers face?				
L4: KT5.4: How did the RAMC and FANY operate the chain of evacuation?				
L5: KT5.5: What incredible medical advances were forged on the Western Front?				
Final Unit Grade:				

L1: KT5.1: What was the historical and medical context of the Western Front?

LEARNING OBJECTIVES

- ☐ Explain the state of surgical asepsis, radiology, and transfusion science on the eve of World War I.

Do Now Activity

Score: / 6

1. What lifestyle choice was definitively linked to lung cancer in the 1950s?

2. Name one modern technology used to diagnose lung cancer.

3. How does the government try to prevent people from smoking?

4. Explain why lung cancer is so difficult to treat effectively.

5. Compare the government's response to lung cancer with its response to cholera in the 19th century.

6. Recall: Who was Edward Jenner and what was their key contribution to medicine?

Vocabulary Check

Fill in the blanks using the vocabulary words below.

Aseptic surgery | Mobile X-ray Unit | Blood Depot

Before the outbreak of the First World War, major breakthroughs in Britain laid the foundation for wartime medical advances. In operating theatres, surgeons transitioned to _____ to prevent microbes from entering wounds during procedures. Meanwhile, the discovery of radiation allowed for the use of _____ designs to find deeply embedded shrapnel in wounded soldiers closer to the battlefield. Finally, to treat severe blood loss and shock, doctors developed methods of preserving and storing blood in a _____ so that transfusions could be carried out quickly without the donor needing to be present.

Q1. Identify the main geographical feature of the Western Front.

Q2. Describe the conditions of the trenches during the war of attrition.

Q3. Explain how artillery bombardment worsened the medical situation.

Q4. To what extent was the RAMC prepared for the medical challenges of the Western Front?

KNOWLEDGE CHECK (Q5)

What was the main limitation of performing blood transfusions in hospital settings before the discovery of sodium citrate in 1915?

- ☐ A. Blood types had not yet been discovered, so transfusions were always fatal.
- ☐ B. Blood could not be stored because it quickly coagulated (clotted), meaning a donor had to be physically present.
- ☐ C. Syringes had not yet been invented.
- ☐ D. The Church banned the transfer of blood between two people.

Pair & Share Activity

Prompt: Which pre-war medical development was the most critical for the survival of soldiers on the Western Front: the rise of aseptic surgery, the invention of mobile X-rays, or the progress in storing and transfusing blood?

Think: Jot down your choice and one piece of evidence from your knowledge (such as the high rate of infection from muddy trench soil making aseptic methods vital, or blood transfusions preventing fatal shock before surgery).

Your Notes:

Q Describe one feature of the First Battle of Ypres. (2 marks)

Q Describe one feature of the Battle of the Somme. (2 marks)

1. Describe one feature of the support trench system on the Western Front. (2 marks)

2. Describe one feature of the problems involved in transporting wounded soldiers away from the battleground. (2 marks)

3. How useful are Sources A and B for an enquiry into the difficulties faced by soldiers in the trench system of the Western Front? Explain your answer, using Sources A and B and your knowledge of the historical context. (8 marks)

Source A

Source A: From the personal pocket diary of Private Thomas Atkins, serving with the British Infantry in the Ypres Salient, November 1916. 'The communication trenches are flooded with three feet of icy water, and the clay walls are collapsing into thick slime. Our boots are permanently waterlogged, and we have to dig out our firesteps constantly. The enemy shelled our support lines yesterday, throwing up colossal geysers of wet mud that choked our drainage channels.'

Source B

Source B: An official photograph taken on the Somme in October 1916. It shows British soldiers attempting to maintain a communication trench.

4. How could you follow up Source A to find out more about the work of the stretcher bearers on the Western Front? (4 marks)

Source A

Source A: A diary entry from an RAMC officer at the Somme, 1916: 'The terrain is devastated. We cannot use the motor ambulances as the heavy rains and artillery have turned the roads into deep quagmires of mud. We are forced to use six horses to pull a single wagon.'

General Notes

This image shows a single sheet of white paper with horizontal blue ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

☐ Emerging (1-2) ☐ Emerging+ (3) ☐ Expected (4-5) ☐ Expected+ (6-7) ☐ Greater Depth (8-9)

Teacher Comment:

L2: KT5.2: How did the trench environment create new medical challenges?

LEARNING OBJECTIVES

- ☐ Explain the causes, symptoms, and preventive strategies developed to combat trench foot and trench fever.
- ☐ Analyze the impact of weaponry on injuries, specifically focusing on shrapnel, head injuries, and gas gangrene from soil.

Do Now Activity

Score: / 6

1. When did the First World War begin and end?

2. What was the 'Western Front'?

3. Name the three major battles fought by the British on the Western Front.

4. Explain the impact of the terrain at Ypres on the soldiers.

5. Why was the casualty rate so high on the Western Front?

6. Recall: Who was Joseph Lister and what was their key contribution to medicine?

Vocabulary Check

Write a historically accurate sentence connecting two terms from the glossary box below.

Trench Foot | Trench Fever | Shell Shock

Your Sentence:

Q1. Identify the cause of Trench Foot.

Q2. Describe the symptoms of Trench Fever.

Q3. Explain how the army attempted to prevent Trench Fever.

Q4. Evaluate the effectiveness of military discipline in preventing environmental illnesses.

KNOWLEDGE CHECK (Q5)

How did the British Army attempt to prevent soldiers from developing trench foot in wet and muddy conditions?

- ☐ A. By providing them with waterproof rubber wellington boots.
- ☐ B. By issuing orders for soldiers to rub their feet with whale oil and regularly change their socks.
- ☐ C. By administering a trench foot vaccine to all soldiers.
- ☐ D. By heating the trenches with underground pipes.

Pair & Share Activity

Prompt: Which aspect of the trench environment presented the most difficult challenge for the Royal Army Medical Corps (RAMC): the physical environment (producing trench foot), the biological vectors (producing trench fever), or the psychological strain of industrial warfare (producing shell shock)?

Think: Jot down your choice and one piece of evidence from the sources, such as the thousands of men evacuated for trench foot, the difficulty of delousing crowded trenches to stop lice, or the fact that shell shock was a completely new condition requiring specialists.

Your Notes:

GCSE Exam Practice

Source A

From the personal memories of William Easton, an 18-year-old volunteer with the East Anglian Field Ambulance, describing the Passchendaele/Ypres campaign in 1917.

"During the third offensive at Ypres, we navigated paths choked with thick, hip-deep mire. In the devastated sector of Hooge, we struggled to relocate casualties. The physical exertion of a single carry left us utterly spent; delivering just two or three wounded men to safety daily was our absolute limit. We worked in teams of four, using shoulder straps to stabilize the stretcher and prevent agonising falls."

Source B

An official archival photograph taken on the Somme in 1916, showing a horse-drawn ambulance navigating a deeply rutted, waterlogged dirt road.



Provenance Scaffolding:

Think about who wrote or produced this source. For Source A, it's an East Anglian Field Ambulance worker at Ypres. Does this personal, on-the-ground experience make it more reliable for understanding the physical toll? For Source B, it's an official photograph from the Somme. Was it meant to document reality, or perhaps carefully framed for morale?

Q6. 2 (a). How useful are Sources A and B for an enquiry into the impact of the terrain on the transport of the wounded on the Western Front? Explain your answer, using Sources A and B and your knowledge of the historical context. (8 marks)

Q Describe one feature of the trench system. (2 marks)

Q Describe one feature of trench foot. (2 marks)

1. Describe one feature of poison gas attacks on the Western Front. (2 marks)

2. Describe one feature of the fighting on the Western Front that led to a high number of injuries. (2 marks)

3. How useful are Sources A and B for an enquiry into the problem of ill health arising from the trench environment? Explain your answer, using Sources A and B and your knowledge of the historical context. (8 marks)

Source A

Source A: A report from a Medical Officer in 1915: 'Altogether about 200 men had to be evacuated from our section of the trenches with swollen, numb feet due to the freezing mud. Several require amputation.'

Source B

Source B: A diary entry from an RAMC surgeon: 'The men are brought in with horrific shrapnel injuries. The metal drives the filthy, manured soil deep into the muscle, creating an environment where gas gangrene thrives. We often have to amputate to save their lives.'

4. How could you follow up Source A to find out more about the problem of trench foot? (4 marks)

Source A

Source A: A report from a Medical Officer in 1915: 'Altogether about 200 men had to be evacuated from our section of the trenches with swollen, numb feet due to the freezing mud. Several require amputation.'

General Notes

This image shows a full page of blank, lined paper. It features approximately 20 horizontal blue lines spaced evenly across the page, typical of notebook paper. The lines are thin and light blue, set against a plain white background. There are no margins, text, or other markings on the page.

- ☐ Emerging (1-2) ☐ Emerging+ (3) ☐ Expected (4-5) ☐ Expected+ (6-7) ☐ Greater Depth (8-9)

Teacher Comment:

L3: KT5.3: What new illnesses and wounds did soldiers face?

LEARNING OBJECTIVES

- ☐ Explain the connection between artillery weaponry, shrapnel, and the introduction of the steel Brodie helmet in 1915.
- ☐ Analyse how the muddy, manured environment of Flanders led to severe wound infections like gas gangrene and tetanus.

Do Now Activity

Score: / 6

1. What was the purpose of the duckboards in the trenches?

2. Name the pattern in which trenches were dug.

3. What was 'No Man's Land'?

4. Explain why the zig-zag pattern of trenches was crucial for survival.

5. To what extent did the trench environment itself cause casualties?

6. Recall: Who was Francis Crick and what was their key contribution to medicine?

Vocabulary Check

Write a clear definition and a historically accurate sentence for the term: **Gas Gangrene**

Q1. Identify the weapon that caused over 60% of all wounds on the Western Front.

Q2. Describe the physical and psychological effects of poison gas.

Q3. Explain why shrapnel wounds almost always became infected.

KNOWLEDGE CHECK (Q4)

Why were deep shrapnel wounds on the Western Front almost guaranteed to develop the deadly infection known as gas gangrene?

- ☐ A. Because the German army dipped their bullets in poison before firing them.
- ☐ B. Because the French farmland had been heavily fertilized with manure for centuries, filling the soil with deadly tetanus and gangrene bacteria.
- ☐ C. Because the British soldiers refused to wash their uniforms.
- ☐ D. Because chlorine gas from chemical attacks infected the wounds.
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Q5. To what extent could doctors successfully treat deep infections during WWI?

Pair & Share Activity

Prompt: Which posed a more difficult medical challenge for the Royal Army Medical Corps (RAMC) on the Western Front: the physical injuries caused by industrial weaponry (such as shrapnel wounds and head trauma) or the environmental infections caused by the Flanders terrain (such as gas gangrene, tetanus, and trench foot)?

Think: Jot down your choice and one piece of evidence from the sources, such as the introduction of the Brodie helmet reducing head wounds vs. the manured soil causing gas gangrene infections that standard antiseptics couldn't treat.

Your Notes:

GCSE Exam Practice

Source A

From a wartime notebook of Lance Sergeant Elmer Cotton in 1915, providing a private, firsthand account of a gas attack.

"A chlorine discharge floods the victim's respiratory tract, effectively suffocating them while standing on dry ground. The resulting symptoms are excruciating: a blinding headache, an overwhelming craving for water—though drinking it brings immediate fatality—and a severe, stabbing sensation in the lungs. Patients continuously expectorate a distinctive greenish fluid. It represents a truly horrific way to perish."

Source B

An official photograph capturing blinded soldiers forming a chain outside an Advanced Dressing Station.



Provenance Scaffolding:

For Source A, consider that it is from a wartime notebook of Lance Sergeant Elmer Cotton in 1915. This is a private, unedited firsthand account. How does this affect its usefulness? For Source B, an official photograph showing blinded soldiers forming a chain. Photographs capture reality but can also be staged. Does the clear visual evidence of the gas's impact make it highly useful regardless?

Q6. 2 (a). How useful are Sources A and B for an enquiry into the effects of poison gas attacks on soldiers on the Western Front? Explain your answer, using Sources A and B and your knowledge of the historical context. (8 marks)

Q Describe one feature of the effects of poison gas. (2 marks)

Q Describe one feature of shrapnel wounds. (2 marks)

1. Describe one feature of aseptic surgery in the early 20th century. (2 marks)

2. Describe one feature of early x-ray machines. (2 marks)

3. How could you follow up Source A to find out more about the treatment and prevention of the effects of gas attacks? (4 marks)

Source A

Source A: From an official report written by a medical officer at an Advanced Dressing Station near Ypres, May 1915: 'At 4:00 AM, the enemy released a dense cloud of yellowish-green gas toward our trenches. Many of the men did not adjust their woolen flannel pads quickly enough. Over fifty casualties were brought into our Dressing Station, their eyes swollen shut and blinded by the vapor. We washed their eyes with bicarbonate solution, but several have already suffocated.'

General Notes

[illegible]

- ☐ Emerging (1-2) ☐ Emerging+ (3) ☐ Expected (4-5) ☐ Expected+ (6-7) ☐ Greater Depth (8-9)

Teacher Comment:

L4: KT5.4: How did the RAMC and FANY operate the chain of evacuation?

LEARNING OBJECTIVES

- ☐ Explain the four main stages of the Western Front chain of evacuation and the mode of transport between them.
- ☐ Evaluate the roles, significance, and contributions of the RAMC and FANY on the Western Front.

Do Now Activity

Score: / 6

1. What is Trench Foot?

2. Which weapons caused the most wounds on the Western Front?

3. Why were wounds highly prone to infection on the Western Front?

4. Explain how gas attacks affected the soldiers.

5. How effective were the preventative measures against gas attacks?

6. Recall: Who was Florence Nightingale and what was their key contribution to medicine?

Vocabulary Check

Fill in the blanks using the vocabulary words below.

Chain of Evacuation | Casualty Clearing Station (CCS) | First Aid Nursing Yeomanry (FANY)

To save soldiers wounded in the trenches, they had to be moved quickly along the _____ before soil-borne bacteria caused fatal infections. While basic first aid was given close to the battlefield, soldiers with life-threatening wounds were sent further back to a _____ for urgent surgery. To transport these casualties safely, the army relied on volunteers from the _____, who braved shellfire to drive motorized ambulances between frontline dressing stations and the rear hospitals.

Q1. Identify the first stage of the Chain of Evacuation.

Q2. Describe the role of Casualty Clearing Stations (CCS).

Q3. Explain how patients were transported between the different stages of the evacuation chain.

Q4. Evaluate the significance of the Chain of Evacuation.

KNOWLEDGE CHECK (Q5)

What was the main difference between the medical treatments performed at a Regimental Aid Post (RAP) and a Casualty Clearing Station (CCS)?

- ☐ A. The RAP performed complex surgery, while the CCS only applied bandages.
- ☐ B. The RAP provided immediate first aid near the frontline, while the CCS was equipped with operating theatres and X-ray machines to perform life-saving surgery.
- ☐ C. The RAP was for officers only, while the CCS was for regular soldiers.
- ☐ D. The RAP was located in Britain, while the CCS was in France.

Pair & Share Activity

Prompt: Was the medical service's success on the Western Front driven more by the highly organized military logistics of the RAMC, or by the courage and adaptability of voluntary organizations like the FANY?

Think: Jot down your choice and one piece of evidence from your knowledge (such as the systematic stages of triage from RAP to Base Hospital vs. FANY volunteers driving motorized ambulances under active bombardment).

Your Notes:

Q Describe one feature of the FANY. (2 marks)

Q Describe one feature of the Chain of Evacuation. (2 marks)

1. Describe one feature of the underground hospital at Arras. (2 marks)

2. Describe one feature of the work of the FANY on the Western Front. (2 marks)

3. How useful are Sources A and B for an enquiry into the work of medical staff in the Casualty Clearing Stations (CCS) on the Western Front? Explain your answer, using Sources A and B and your knowledge of the historical context. (8 marks)

Source A

Source A: A letter from an RAMC nurse working at a Casualty Clearing Station (CCS) in 1916: 'We worked night after night, in the thunderous noise of raging battles. The surgeons are constantly amputating limbs to stop the spread of gangrene before putting the men on the trains.'

Source B

Source B: A diary entry from a member of the FANY in 1917: 'I drove my motor ambulance through the terrible mud for 14 hours straight today, bringing men from the Main Dressing Station down to the railhead.'

4. How could you follow up Source A to find out more about the work of medical staff in the Casualty Clearing Stations? (4 marks)

Source A

Source A: A letter from an RAMC nurse working at a Casualty Clearing Station (CCS) in 1916: 'We worked night after night, in the thunderous noise of raging battles. The surgeons are constantly amputating limbs to stop the spread of gangrene before putting the men on the trains.'

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Teacher Comment:

L5: KT5.5: What incredible medical advances were forged on the Western Front?

LEARNING OBJECTIVES

- ☐ Explain how the Thomas splint and mobile X-ray units solved critical evacuation and surgical challenges.
- ☐ Analyze the development of blood transfusion and the significance of the first blood bank at the Battle of Cambrai.

Do Now Activity

Score: / 6

1. What does RAMC stand for?

2. What was the role of the FANY?

3. Describe the 'Chain of Evacuation'.

4. Explain the importance of the Casualty Clearing Stations (CCS).

5. Why was the work of the RAMC so dangerous?

6. Recall: Who was James Simpson and what was their key contribution to medicine?

Vocabulary Check

Write a historically accurate sentence connecting two terms from the glossary box below.

Thomas Splint | Carrel-Dakin Method | Blood Depot

Your Sentence:

Q1. Identify the device designed by Hugh Owen Thomas.

Q2. Describe how Richard Lewisohn improved blood transfusions.

KNOWLEDGE CHECK (Q3)

How did the implementation of the Thomas splint on the Western Front in 1916 directly reduce mortality rates among wounded soldiers?

- ☐ A. It stopped the spread of infection by killing the bacteria that caused gas gangrene.
- ☐ B. It cured shell shock by holding the soldier firmly in place.
- ☐ C. It kept fractured leg bones rigidly in place during transport, preventing fatal blood loss and shock.
- ☐ D. It provided a steady drip of blood for transfusions on the battlefield.

Q4. Explain how the Thomas Splint improved survival rates.

Q5. To what extent did the Western Front accelerate medical progress?

Pair & Share Activity

Prompt: Which medical advance on the Western Front represented a more impressive leap forward: the mechanical simplicity of the Thomas Splint which stabilized fractures, or the complex chemical engineering of blood preservation and storage?

Think: Jot down your choice and one piece of evidence from the sources (e.g., the Thomas splint keeping the leg rigid to prevent internal bleeding during transport, or sodium citrate allowing blood to be stored in depots for the first time without donors present).

Your Notes:

GCSE Exam Practice

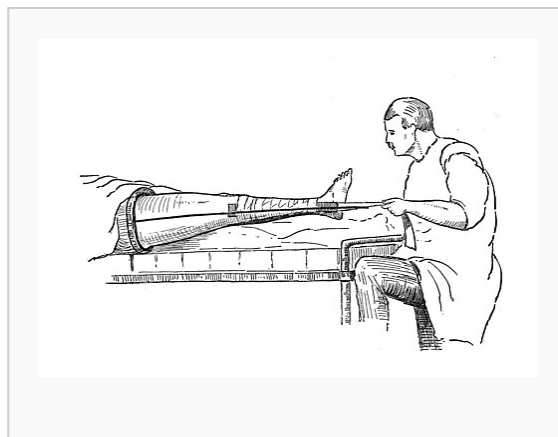
Source A

From an official clinical report written by Captain Arthur Vaughan, a surgeon serving with the Royal Army Medical Corps (RAMC) at a Casualty Clearing Station (CCS) behind the frontline, October 1916.

"Our medical team at the Casualty Clearing Station has fully adopted the new leg traction splint. Before we received this iron frame, almost every compound femur fracture arrived at our ward in a state of terminal shock due to the jagged bones grating inside the flesh during transit. Since implementing this immobilization frame last month, we have kept the fractured limbs completely rigid. Out of forty admissions with thigh wounds, we have lost only six patients."

Source B

A studio photograph taken to train medical orderlies, showing a soldier's leg secured in a Thomas splint.



Provenance Scaffolding:

Source A is a clinical report by a surgeon; does this make its statistics reliable? Source B is a staged training photograph; while not taken on the chaotic frontline, does this actually make it better for demonstrating how the device technically worked?

Q6. 2 (a). How useful are Sources A and B for an enquiry into the effectiveness of new medical developments used to treat wounded soldiers on the Western Front? Explain your answer, using Sources A and B and your knowledge of the historical context. (8 marks)

1. Describe one feature of the Thomas Splint. (2 marks)

2. Describe one feature of the medical advances in blood transfusions on the Western Front. (2 marks)

3. How useful are Sources A and B for an enquiry into the effectiveness of new medical developments used to treat wounded soldiers on the Western Front? Explain your answer, using Sources A and B and your knowledge of the historical context. (8 marks)

Source A

Source A: From an official clinical report written by Captain Arthur Vaughan, a surgeon serving with the Royal Army Medical Corps (RAMC) at a Casualty Clearing Station (CCS) behind the frontline, October 1916.

'Our medical team at the Casualty Clearing Station has fully adopted the new leg traction splint. Before we received this iron frame, almost every compound femur fracture arrived at our ward in a state of terminal shock due to the jagged bones grating inside the flesh during transit. Since implementing this immobilization frame last month, we have kept the fractured limbs completely rigid. Out of forty admissions with thigh wounds, we have lost only six patients.'

Source B

Source B: A photograph from a British medical instruction manual, published in 1917, showing the deployment of a traction device.

(Visual Description: A detailed, black-and-white studio photograph showing a soldier lying flat on a wooden stretcher. His right leg is suspended and held rigid inside a metallic Thomas splint, which consists of a padded leather ring pushed up against the groin and parallel steel rods running down both sides of the limb. Bandages and canvas straps are wrapped tightly around the thigh and calf to keep the leg completely straight. A canvas sling is hooked to the end of the splint to pull the foot forward, creating traction. Two medical orderlies in clean aprons

stand alongside, demonstrating how to attach the secure traction cords.)

4. How could you follow up Source A to find out more about new techniques being used on the Western Front to deal with injuries? (4 marks)

Source A

Source A: A report from a surgeon at a CCS in 1916: 'Several unsuccessful trials this morning to extract a shell fragment from a soldier's brain. Finally, we decided to try using a large wire nail as a magnet.'

General Notes

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- ☐ Emerging (1-2) ☐ Emerging+ (3) ☐ Expected (4-5) ☐ Expected+ (6-7) ☐ Greater Depth (8-9)

Teacher Comment:

Factors Overview: Western Front

Edexcel focuses heavily on the factors that drove medical progress (or held it back). For each factor below, write one specific historical example from this period that either helped or hindered medical progress.

Factor	Specific Historical Example & Impact
Individuals	
The Church & Religion	
Government & Wealth	
Science & Technology	
Attitudes in Society	
War	

Factor	Specific Historical Example & Impact